HurricaneReliefFacade

-instance: HurricaneReliefFacade (static)
-users: UserList
-shelters: ShelterList
-requests: ReliefRequestList
-gps: gpsClass
-currentUser: User
-Hurricane: hurricane

-HurricaneReliefFacade()
+getInstance() : HurricaneReliefFacade
+login(String email, String password): boolean
+logout(): void
+signUp(String firstName, String lastName, int age, String phonenumber, String email, String password, Languages language, String currentLocation):boolean
+submitEmergencyRequest(UUID victimId, String location, String aidType, String urgencyLevel, String specialNeeds, boolean petNeeds): Relief Request
+submitPetAssistedRelief(UUID victimId, String location, String aidType, Pet pet) : ReliefRequest
+updateUrgencyLevel(UUID requestId, String level) : boolean
+ cancelReliefRequest(UUID requestId) : boolean
+assignVolunteerToTask(UUID volunteerId, UUID requestId): void
+dispatchReliefRoute(UUID volunteerId, String start, String destination) : void
+completeAssignedTask(UUID volunteerId, UUID requestId): void
+checkInVictim(UUID shelterId, UUID victimId): void
+updateSupplyInventory(UUID shelterId, int waterCount, String foodSupplies) : void
+requestMedicalSupport(UUID shelterId, String supportType) : void
+getShelterStatusReport(UUID shelterId): String

ShelterList

- shelters: ArrayList<Shelter>

+shelterList()
+ addShelter(Shelter shelter): void
+ removeShelter(UUID shelterId): boolean
+ getShelterById(UUID shelterId): Shelter
+ getAvailableShelters(): ArrayList<Shelter>
+ getPetFriendlyShelters(): ArrayList<Shelter>

Shelter

- location: String
- ShelterId: UUID
- capacity: int
- occupancy: int
- status: enum
- petFriendly: boolean

+ shelterStatus()
+getAvailableSpace()
+getMedInfo()
+getReligiousServ()
+getProtocol()
+shelterFlag()

Resources

+ foodSupplies: ArrayList<String>
+ waterSupplies: int
+ medicineSupplies: ArrayList<String>
+ otherSupplies: ArrayList<String>

+ getFoodSupplies(): ArrayList<String>
+getWaterSupplies(): ArrayList<String>
+ updateSupplies():
+resourcesAlert()

Hurricane

- name: String
- category: int
- windSpeed: double
- affectedZipCodes: ArrayList<String>
- location: String
- affectedAreas: ArrayList<String>
- condition: String
- terrain: String

+ getUpdates(): String
+ trackPath(): void
+ getCategory(): int
+ updateWindSpeed(double newSpeed): void
+ getAffectedAreas(): ArrayList<String>
+ getCurrentLocation()
+ findAffectedAreas()

gpsClass

- location: *String*
- shelterAddress: ArrayList<String>
- affectedAreas: ArrayList<String>
- satelliteCon: boolean
- disabilityCheck: boolean

+ disableRoute()
+ volunteerCheck(Volunteer v)
+ volunteerRoute(Volunteer v)
+ callEMS()
+ calculateRoute(String start, String destination)
+downloadOfflineMap()
+ getAccessibleRoute()

User

- firstName: String
- lastName: String
- age: int
- UserId: UUID
- password:String
- language: enum
- currentLocation: String
-admin: boolean
-phoneNumber: String
-email: String

+User(String firstName, String lastName, int age, String phoneNumber, String email, String password, Languages language, String currentLocation)
+User(UUID userId, String firstName, String lastName, int age, String phoneNumber, String email, String password, Languages language, String currentLocation, boolean admin)
+ authenticate(String password): boolean
+ createAccount(): void
+ login (String password): boolean
+ selectLanguage(Language language):void
+ setAccessibility(): void
+ resetPassword(String newPassword):void
+logout(): void
+deleteAccount(): void
+verifyPhone(): boolean
+isAdmin(): boolean
+getEmail() : String
+getPhoneNumber(): String

<<enumeration>> Languages

ENGLISH
SPANISH
FRENCH

UserList

- users: ArrayList<User>

+addUser(User user): void
+addUser(String firstName, String lastName, int age, String phoneNumber, String email, String password, enum language, String currentLocation) : void
+ removeUser(UUID userId): boolean
+ getUserById(UUID userId): User
+ getVolunteers(): ArrayList<Volunteer>
+ getVictims(): ArrayList<Victim>
+getUserByEmail(String email): User

DataLoader

+loadUsers() : ArrayList<User>
+loadShelters(): ArrayList<Shelter>
+loadRequests(): ArrayList<ReliefRequest>

DataWriter

+saveUsers(ArrayList<User> users) : void
+saveShelters(ArrayList<Shelter> shelters): void
+saveRequests(ArrayList<ReliefRequest> requests): void

Victim (extends user)

-guest: ArrayList<String>
-member: ArrayList<String>
-pet: pet
-safe: boolean

+Victim(String firstName, String lastName, int age, String phoneNumber, String email, String password, Languages language, String currentLocation)
+Victim(UUID id, String firstName, String lastName, int age, String phoneNumber, String email, String password, Languages language, String currentLocation, boolean admin, ArrayList<String> guest, ArrayList<String> member, Pet pet, boolean safe)
+ createReliefRequest(String victimName, String location, String, aidType, String urgencyLevel,)
+editRequest(ReliefRequest r)
+deleteRequest(ReliefRequest r)
+viewRequestStatus(ReliefRequestr)
+requestVolunteer()
+viewRequestHistory()
+rateExperience()
+markSafe()

Volunteer

- backgroundCheck: Boolean
- assignedSkills: ArrayList<Skill>
- certifications: ArrayList<String>

+viewMap()
+ claimRequest(ReliefRequest r)
+viewEtaToRequest(ReliefRequest r)
+ updateInventory(String item, int quantity)
+ getVictimStatus(Victim v)
+markUnavailable()
+completeRequest()

Pet

- type: String
- specialNeeds: String
- medicationRequired: boolean
- mobilityIssues: boolean

+getType(): String
+getSpecialNeeds(): String
+isMedicationRequired(): boolean
+isMobilityIssues(): boolean

<<enumeration>> Skill

VET
FIRST_AID
EVACUATION

ReliefRequestList

-request: ArrayList<ReliefRequest>

+addRequest(ReliefRequest request) : void
+addRequest(String victimName, String location, AidType aidType, String specialRequests, UrgencyLevel urgencyLevel, boolean petNeeds) : void
+ removeRequest(String requestId): boolean
+getRequestByid(String requestId) : ReliefRequest
+getRequestsByUrgency(UrgencyLevel level) : ArrayList<ReliefRequest>
+getPendingRequests(): ArrayList<ReliefRequest>

ReliefRequest

- requestId: UUID
- user: User
- victimName: String
- location: array
- aidType: array
- specialRequests: String
- urgencyLevel: array
- status: String
- petNeeds: bool

+ReliefRequest(User user, String victimName, enum location, enum aidType, String specialRequests, enum urgencyLevel, bool petNeeds)
+ReliefRequest(UUID requestId User user, String victimName, enum location, enum aidType, String specialRequests, enum urgencyLevel, bool petNeeds, String status)
+ updateStatus(urgencyLevel string level, status String level)
+ callEMS ()
+attachPhoto(String photoPath)
+viewETA()